MIL-C-46942 (MU)
22 JULY 1963
SUPERSEDING
FA-PD-MI-2215, Rev. 2
3 MARCH 1959

MILITARY SPECIFICATION

CATAPULT, AIRCRAFT EJECTION SEAT, M5A1 ASSEMBLY

1. SCOPE

1.1 This specification covers Catapult, Aircraft Ejection Seat, M5A1, Assembly, loaded with Cartridge, Aircraft Ejection Seat, M28A1, for use in conjunction with other propellant actuated devices in air-crew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for

MIL-I-45208 — Inspection Requirements, General Specification for

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 — Quality Assurance Terms and Definitions

MIL-STD-414 — Sampling Procedures and Tables for Inspection by Variables for Percent Defective

MS-28741 — Hose Assembly-Detachable End Fitting, Medium Pressure

DRAWINGS

MUNITIONS COMMAND

F8593686 — Catapult, Aircraft Ejection Seat, M5A1, Assembly

F8797377 — Ammunition, Container, Fiber M299 for Catapult, Aircraft Ejection Seat, M5A1

F8797378 — Box, Packing, Ammunition, for Catapult, Aircraft Ejection Seat, M5A1

IEL-8593686 — Index to Inspection Equipment List for Catapult, Aircraft Ejection Seat, M5A1, Assembly

FSC 1375

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

INTERSTATE COMMERCE COMMISSION — 49 CFR 71–78 — The Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles. (The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71–78.").

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with all the requirements specified on the applicable drawings and specifications.

3.2 Torque requirements. Torque requirements shall be as specified in Table I. No movement or deformation shall occur upon application of the breakaway torque.

TABLE I

Components	*Seating or bottoming (inch pounds)*	*Breakaway (inch pounds) (minimum)*
Block and Inside Tube after insertion of Cartridge	400 ± 50	300
Stop Ring and Trunnion	500 ± 50	400

3.3 Assembly.

3.3.1 The flange face surface of the plug shall be in complete contact with the cartridge.

3.3.2 The block shall be free to rotate 360 degrees in the counterclockwise direction.

3.4 Residual magnetism. The catapult shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.5 Functional.

3.5.1 *Ballistics.* Catapult loaded with accepted M28A1 cartridges shall propel a 300 pound mass vertically and shall comply with the requirements listed in Table II (see 6.3).

TABLE II

Temperature	*Velocity (min.) (see 6.3)*	*Acceleration, pressure time*	*Max. load cell*	*Rate of change of acceleration (max.)*	*Ignition delay (max.) (see 6.2)*
—65° to —70°F	48 fps	16g	14g	100g/sec	150 milliseconds
65° to 75°F	60 fps	20g	18g	170g/sec	150 milliseconds
160° to 165°F	64 fps	24g	24g	250g/sec	150 milliseconds

3.5.2 The catapult shall not fail to fire.

3.5.3 The catapult shall function without mechanical failure (see 6.1).

3.6 Workmanship. The requirements for workmanship are as specified on the applicable drawings, specifications and the following:

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3.6.1 *Metal defect.* All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.6.2 *Burr.* No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, rust, corrosion or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to Standard MIL–STD–109 to define the quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with Specification MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 Inspection provisions.

4.2.1 *Lot.*

4.2.1.1 *Presentation of lots.* **The product** shall be presented in accordance with Standard MIL–STD–105 or Standard MIL–STD–414, as applicable.

4.2.1.2 *Size of lot.* A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table III. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table III. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE III

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1 – 300	100% examination	.....	.....	.....	.....
301 – 800	50	0	1	1	2
801 – 1800	90	1	2	2	3

4.2.2.1 *Cartridge, Aircraft Ejection Seat Catapult, M28A1, Dwg. D8593353.*

Defect	Method of inspection
Critical:	
1. Mixed ammunition	Visual
2. Case split or perforated	Visual
3. "O" ring missing	Visual
4. Sealing disc missing or damaged	Visual
5. Marking missing, incorrect or illegible	Visual

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Defect	Method of inspection
Major:	
101. "O" ring split, cut or cracked	Visual
102. Foreign matter	Visual
Minor:	
201. Case scratched, wrinkled or bulged	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 *Seal, Dwg. C8593665.*

Defect	Method of inspection
Critical:	
None defined.	
Major:	
None defined.	
Minor:	
201. Inside diameter under flange	Gage
202. Inside diameter adjacent to flange	Gage
203. Radius missing or incorrect	Visual
204. Metal defective	Visual
205. Protective finish inadequate or defective	Visual
206. Burr	Visual
207. Foreign matter	Visual

4.2.2.3 *Assembly, Dwg. F8593686, less all drawings except C8593643, C8593644, F8593648, B8593654, C8593656, C8593657, and AN 6227–11.*

Defect	Method of inspection
Critical:	
1. Gas port hole obstructed	Visual
2. Any part missing	Visual
3. Improper assembly	Visual
Major:	
101. Thread damaged	Visual
102. Burr	Visual
103. Foreign matter	Visual
Minor:	
None defined.	

4.2.2.4 *Assembly, Dwg. F8593686, less drawings B8593660, C8593691, D8593693, C8593699 and D8593700.*

Defect	Method of inspection
Critical:	
1. Flange face surface of plug not in complete contact with cartridge	Gage
2. Cartridge missing	Visual
3. Seals on inside tube missing	Visual
Major:	
101. Shock washer missing	Visual
102. Burr	Visual

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Defect	Method of inspection
103. Foreign matter	Visual
Minor:	
None.	

4.2.2.5 *Assembly, Dwg. F8593686, less all drawings except B8593660, C8593691, D8593693, C8593699 and D8593700.*

Defect	Method of inspection
Critical:	
1. Seal on trunnion missing	Manual/Visual
Major:	
101. Burr	Visual
102. Foreign matter	Visual
Minor:	
None. defined.	

4.2.2.6 *Assembly, Dwg. F8593686 (with Stop Ring not engaged with Trunnion).*

Defect	Method of inspection
Critical:	
1. Tubes not free to slide	Manual
Major:	
None defined.	
Minor:	
None defined.	

4.2.2.7 *Assembly, Dwg. F8593686.*

Defect	Method of inspection
Critical:	
1. Mixed catapults	Visual
2. Marking missing, incorrect or illegible	Visual
Major:	
101. Total length, max	Gage
102. Diameter for mounting hole	Gage
103. Length from cap to mounting hole	Gage
104. Length between mounting points	Gage
105. Block not free to rotate 360 degrees	Manual
106. Burr	Visual
107. Foreign matter	Visual
Minor:	
201. Location of labels, incorrect	Visual
202. Shipping plug missing	Visual
203. Ordnance seal missing or insecure	Visual
204. Protective finish inadequate or defective	Visual

4.2.2.8 *Ammunition, Container, Fiber, Dwg. F8597377.*

Defect	Method of inspection
Critical:	
None defined.	

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Defect	Method of inspection
Major:	
None defined.	
Minor:	
201. Gap between body and cover, max	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.9 *Box, Packing, Ammunition, Dwg. F8797378.*

Defect	Method of inspection
Critical:	
None defined.	
Major:	
None defined.	
Minor:	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual/Manual
203. Strapping missing, loose, broken or improperly assembled	Visual/Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking misleading or unidentifiable	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 *Inspection testing.* The following tests shall be performed in accordance with 4.3, except as provided in Specification MIL–I–45208.

4.2.3.1 *Breakaway torque.* A breakaway torque shall be applied to each catapult threaded joint listed in 3.2. Failure of any unit to comply with the requirements of Table I shall be cause for rejection of the catapult.

4.2.3.2 *Residual magnetism.* Each catapult shall be tested for excessive residual magnetism. Failure of any catapult to comply with the requirements of 3.4 shall be cause for rejection of the catapult.

4.2.3.3 *Functional.* The following test shall be performed by the Government:

4.2.3.3.1 *Ballistics.* Fourteen units shall be tested to determine compliance with the requirements of 3.5.1, 3.5.2 and 3.5.3. The units shall be temperature conditioned as provided in Table IV and 4.3.3.1.1. The variables sampling plan of MIL–STD–414, Example C–2, for AQL listed in Table IV shall be used for each sample of 10 units. Failure of any unit to comply with the requirements of 3.5.1, 3.5.2 and 3.5.3 or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

TABLE IV

Temperature	Sample size	AQL
—65° to —70° F	2	0.65%
65° to 70° F	10	0.65%
160° to 165° F	2	0.65%

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4.2.4 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL–8593686, except as provided in Specification MIL–I–45208.

4.3 Test methods and procedures.

4.3.1 *Breakaway torque.* A torque wrench set at the torque requirement specified in 3.2 shall be used to determine the adequacy of the joints. The assembly shall be secured in the fixture prior to application of the torque.

4.3.2 *Residual magnetism.* The catapult shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the catapult 6 inches from, and at the same height as, the compass in a north-south horizontal position (trunnion end at south) with the compass set in an east-west heading as shown in Figure 1. The catapult shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass.

4.3.3 *Functional.*

4.3.3.1 *Ballistics.* The catapult shall be rigidly supported in the fixture by the mounting points and fired with an M3A1 Initiator connected to the catapult with 10 feet of aircraft hose, size 4, as specified in Standard MS–28741. The catapult shall comply with the requirements of 3.5. The catapult shall be tested within 10 minutes after removal from the conditioning chamber. When this time limit is exceeded, the catapult shall be reconditioned at the original temperature.

4.3.3.1.1 *Temperature conditioning.* A standard M5A1 Catapult, loaded with a dummy M28A1 Cartridge containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. (Level A.) Packing and marking shall be in accordance with Drawings F8797377 and F8797378.

5.2 Delivery. Delivery shall be in accordance with Interstate Commerce Commission Regulations cited in 2.2.

5.3 Data cards. Data card information shall be as specified in Specification MIL–A–2550.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 "Ignition delay" is defined as the time that elapses between the firing of the cartridge primer and the beginning of the continuous pressure rise on the thrust time record.

6.3 Velocity applies at tube separation.

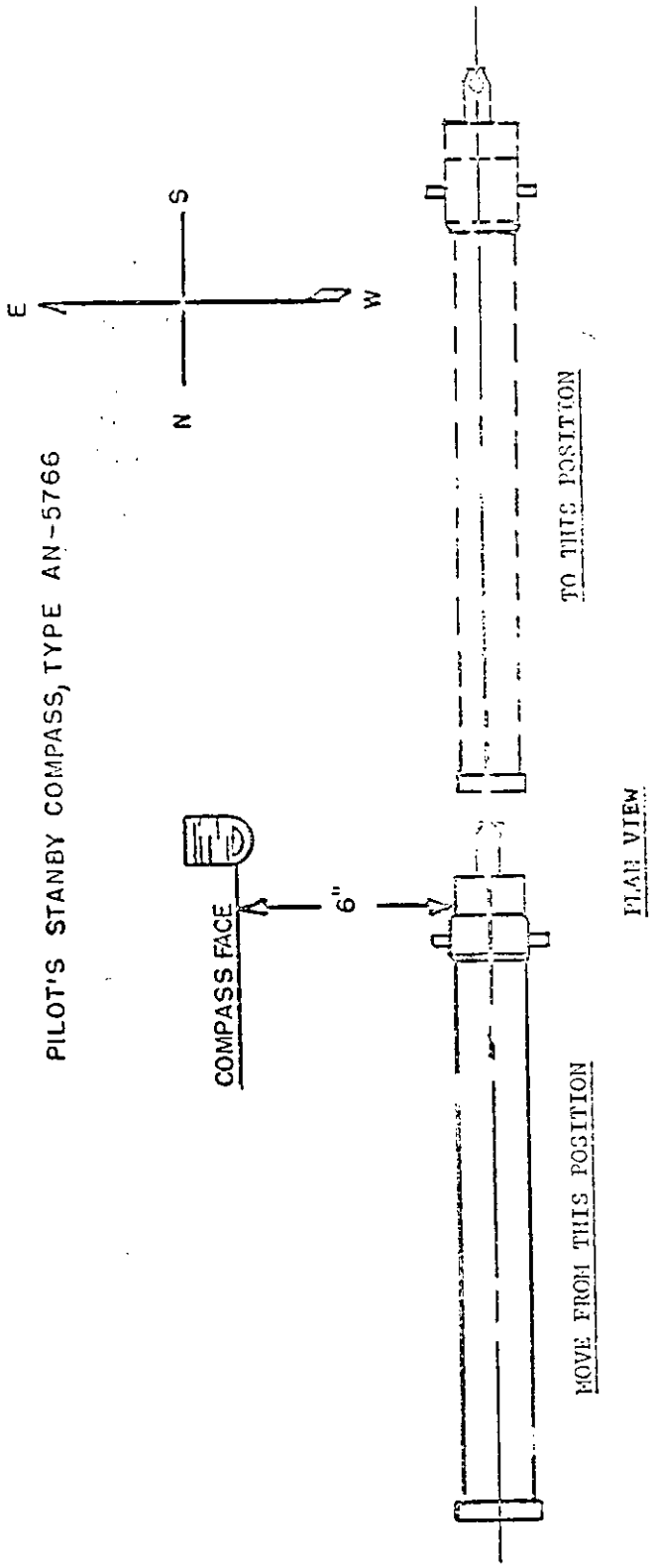
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NOTE:-

1. CATAPULT AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

FIGURE 1. *Layout for determination of residual magnetism in catapult, aircraft ejection seat, MEAL.*

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